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Source: Transactions of the Kansas Academy of Science, 122(3-4) : 235-242

Published By: Kansas Academy of Science

URL: https://doi.org/10.1660/062.122.0306

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Transactions of the Kansas Academy of Science

Vol. 122, no. 3-4
p. 235-242 (2019)

Does activity of white-tailed deer (*Odocoileus virginianus*) vary between human-hunted areas and refugia?

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Hunting by humans impacts behaviors of targeted species, similar to what is observed in other predator-prey systems. Managed wildlife areas commonly have refuges where hunting is prohibited. Deer (family Cervidae) are commonly hunted species and their behavior might differ between such refuges and areas outside refuges where deer hunting is permitted. We predicted that white-tailed deer detections would be higher in refuges than in hunted areas as populations seek refugia from hunting, and that detections would be greatest at night in hunted vs. refuge areas. Using camera traps in refuge and hunted areas on two public wildlife areas in eastern Kansas, we calculated detections of individual deer per hour and categorized days into four time periods (dawn, day, dusk, night). Detection rates of deer overall (sexes pooled) did not vary significantly between refuges and hunted areas. Unexpectedly, male detection rates were higher in hunted areas than refuges. Differences in deer detection rates among time periods were marginally significant for deer overall with detection rates being higher during dawn than daytime, apparently driven by activity of females (as males showed no significant pattern). Sizes of the refuges in our study might not have been large enough to negate the effects of hunting pressure on deer behavior. Despite standardization of habitat characteristics, habitat differences between refuge and hunted areas may have also affected deer behavior. It is also possible that hunting pressure during our study (18 October – 15 November 2018) was not high enough to alter deer behavior. Future study is warranted to account for these variables.

Keywords: Predator-prey, wildlife, Cervidae, refuge, hunting, targeted, behavior

Introduction

Predator species affect the anti-predatory behavior of their prey (Sih 1984; Creel et al. 2005; Gude et al. 2006). Such patterns have been studied extensively within a wide range of biological systems (Lima and Dill 1990; Brown, Laundre and Gurung 1999; Gude et al. 2006; Bonnot et al. 2012; Blumstein 2014). When subjected to predation risk, prey might choose to forage within secondary habitats containing fewer resources than primary habitats and may also alter their temporal patterns of activity in order to avoid potential predators (Lima and Dill 1990; Brown, Laundre and Gurung 1999; Tambling et al. 2015). The behavioral responses of prey to predation risk may generate a cascade of biological consequences potentially leading to decreased fitness in individuals under the threat of predation (Ydenberg and Dill 1986).

In nature, humans can apparently pose a probable threat to other animals which produces alterations in the behavior of animals in contact with human populations (Geist et al. 2005; Bonnot et al. 2012). However, some species become habituated to non-predatory behavior of humans (Blumstein 2014). Nonetheless, most species apparently view humans as a potential threat and respond in ways similar to reactions to other predators (Autry 1967; Wearing and Neil 1999; Christ et al. 2003; de Boer et al. 2004). More specifically, animals have the capacity to learn and associate human hunters with predation. While hunting, humans may

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create stimuli which alert target animals of their presence (e.g., sound, odor, visual encounters, fleeing conspecifics; Griffin 2004) that prey might then associate with threat of predation (Griffin 2004).

Within certain zones of state and federally managed wildlife areas, hunting is not allowed. Individuals found within these refuge zones might therefore behave differently than individuals found in areas open to hunting. Tran et al. (2016) found that fishes in areas open to fishing initiate flight at a greater distance when approached by humans than in marine protected areas where fishing is not allowed. During the hunting season, waterfowl have been found to be more abundant in refugia from hunting than areas where hunting is allowed (Evans and Day 2002).

Human interaction with ungulate mammals (e.g., deer, family Cervidae) induces various changes in ungulate behavior (Stankowich 2008). However, the behavioral responses of ungulates to human presence varies depending upon whether the ungulate species in question is hunted or not (Stankowich 2008). Within human-hunted ungulate species, individuals have been found to be more fearful of human disturbance at dawn and dusk than other times of the day due to increased hunting pressure at these times (Stankowich 2008). Stankowich (2008) found no apparent effect of time of day on the fearfulness of human disturbance in non-hunted ungulate populations. Furthermore, in areas with hunting activity, ungulates had greater flight responses (fled at a greater distance of approach by humans) than in areas with no hunting (Behrend and Lubeck 1968; de Boer et al. 2004; Stankowich 2008). Proffitt et al. (2009) found that elk (*Cervus elaphus*) responded to human hunting pressure in ways that were comparable to the behavioral responses of elk to predation risk from wolves (*Canis lupus*). Elk have also been found to initiate flight at a greater distance of approach after having been exposed to hunting pressure than before being exposed to hunting (Bender, Beyer and Haufler 1999).

White-tailed deer (*Odocoileus virginianus*) range from Northern Canada into South America and are frequently targeted as a game species by human hunters throughout North America. White-tailed deer in refuge areas behave differently than those in hunted areas, as is characteristic of many ungulate species (Behrend and Lubeck 1968; Stankowich 2008). When in areas of refuge, white-tailed deer initiate flight at a shorter distance of approach by humans than within areas of higher hunting pressure (Behrend and Lubeck 1968). During the 2017-2018 hunting season in Kansas, over 150,000 permits were sold for the harvest of deer and approximately 80,000 deer were harvested (Jaster 2018). In Kansas, the hunting season for white-tailed deer is held from mid-September until mid-January with methods of take ranging from archery to rifle (Kansas Department of Wildlife, Parks, and Tourism 2018). Legal shooting hours occur from thirty minutes before sunrise to thirty minutes after sunset (Kansas Department of Wildlife, Parks, and Tourism 2018). White-tailed deer have access to refuge areas in Kansas on both state-managed wildlife areas and National Wildlife Refuges.

Does the presence or absence of hunting pressure cause changes in overall white-tailed deer activity and diel patterns of activity inside versus outside of refuge areas in Kansas? We hypothesized that due to increased hunting pressure within hunted areas, white-tailed deer will be less active and more nocturnal than in refuge areas in an attempt to avoid human hunting pressure, which occurs from thirty minutes before sunrise to thirty minutes after sunset. The objectives of our study were to (1) compare the overall activity and (2) time periods of activity in white-tailed deer (dawn, day, dusk, and night) inside and outside of refuge areas in eastern Kansas. Here we consider "activity," as assessed using remotely triggered cameras, to be proportionate to deer abundance and movements of individual deer that were detected repeatedly over time (though, as we explain below, the latter was less common).

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Methods

Study Areas: The areas within which our study was conducted were the Flint Hills National Wildlife Refuge (FHNWR) and the Melvern Wildlife Area (MWA), both located in east-central Kansas and separated by approximately 16 km. Both areas are managed for the benefit of migratory birds, as well as to improve habitat for many other species of wildlife. White-tailed deer are abundant within both areas (DMO, pers. observ.) and are harvested by humans during the hunting season (September-January) in designated hunting areas. During our study (18 October-15 November), the only legal method of take was archery. However, prior to our study, legal methods also included muzzleloader (17-30 September) and rifle (6-8 October). Within each area, a subset of the area is off limits to hunting either during the deer hunting season, or year-round. These areas are considered "refuges" for migratory and non-migratory species. The FHNWR, covering 7472 ha (2500 ha as refuge), was established in 1966 and is managed by the US Fish and Wildlife Service (USFWS). Habitats found within the FHNWR include wetlands, agricultural lands, grasslands, and riparian woodlands. The riparian woodlands include species such as bur oak (*Quercus macrocarpa*), pecan (*Carya illinoinensis*), black walnut (*Juglans nigra*), American elm (*Ulmus americana*), hackberry (*Celtis occidentalis*), green ash (*Fraxinus pennsylvanica*), and Kentucky coffeetree (*Gymnocladus dioicus*). The MWA, covering 4087 ha (520 ha as refuge), is managed by the Kansas Department of Wildlife Parks and Tourism. Habitats found within the MWA include, grasslands, riparian woodlands, and wetlands. Riparian woodlands are confined to areas near the Marais des Cygnes River and various other streams that flow through the area. Common tree species in the area include American elm, cottonwood (*Populus sp.*), sycamore (*Platanus occidentalis*), and honey locust (*Gleditsia triacanthos*).

Sampling Technique: Sampling was conducted using motion-sensitive camera traps within two treatments (refuge and hunted areas), at four locations (sampling units) per treatment, at each of the two areas (FHNWR and MWA). After sampling for two weeks (18-31 October 2018) each camera was moved to a new location for another two weeks (1-15 November 2018), resulting in sixteen unique camera locations. Each camera location was within a forested area greater than 0.5 ha, within 500 m of a major body of water (river, large creek, or lake), and within 1000 m of an agricultural field. Approximate camera locations were determined, prior to in-field camera establishment, by using aerial imagery from Google Earth (https://www.google.com/earth/, Google LLC, Mountain View CA). Camera traps (Reconyx PC900 Hyperfire Professional IR; Reconyx Inc., Holmen, WI; set for Medium High sensitivity, 3 image bursts, 1 s delay) were installed in areas of similar habitat in both refuges and hunted areas and were secured to trees about 1.5 m above the ground using cable locks. One camera malfunctioned early in trial resulting in a total of 15 unique camera locations (8 within hunted areas, 7 within refuges; 3 in refuges at MWA). Cameras were pointed to the north to avoid images being washed out by sunlight prevailing from the south. Cameras were positioned so that an area of approximately 10 m x 10 m was free of large obstructions (trees, fallen logs, shrubs) in front of the cameras to avoid difficulty in identifying individuals. Cameras were installed at midday in order to minimize disturbance to deer in the area.

SD data cards were removed and replaced as each camera trap was moved to a new location within the same treatment. On 15 November 2018 all camera traps were collected, SD cards were removed, and image data containing white-tailed deer was collected from the SD cards using a laptop computer. Time-period of detections was categorized as dawn (one hour before sunrise to one hour after sunrise), day (between dawn and dusk), dusk (one hour

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before sunset to one hour after sunset), and night (between dusk and dawn). Individuals that could be identified based upon morphology (antler configuration in males, pelage and other morphological distinctions in unantlered deer) were counted only once per each time period during each day. Individuals that could not be uniquely identified based upon morphology were counted each time they were detected, which likely resulted in double-counting of individuals. However, the latter nonetheless constituted an index of deer activity.

Statistical Analysis: Detection rates of numbers of unique individuals per hour were calculated within each time period for each camera. Means of these detection rates, per time period, per camera location, were used as replicates in our analyses. A two-way ANOVA was used to test for relationships between the mean number of detections per hour in relation to treatment (refuge vs. non-refuge) and time period as well as a treatment-by-time interaction ($\alpha = 0.05$). Tukey's HSD post hoc test was used to compare means among the four time periods. Analyses were run with sexes pooled, as well as per each sex independently. All analyses were run in R-studio (version 3.5.1, R Core Team 2013).

Results

In total, 297 detections of deer (unique individuals or repeat activity of unidentifiable individuals) were recorded throughout the study. Detections per camera ranged from 0 to 49 with a mean of 19.8 ± 3.62 SE detections/camera. One-hundred and thirteen unique detections of males and 159 detections of females were recorded during the study. We were not able to determine sex or individual identity for 25 detections (8.4% of total detections). Also, we were unable to uniquely identify 78 detections of females (26.3% of total detections). In total, 103 detections were unable to be uniquely identified (34.7% of total detections).

Detections overall (irrespective of sex), did not vary significantly between refuge ($\bar{x}$= 0.055 ± 0.018 SE detections/camera/hour) and hunted areas ($\bar{x}$= 0.062 ± 0.013 SE detections/camera) (F= 0.564; df= 1, 52; P= 0.456) (Fig. 1A). Differences in deer detections among time periods were marginally significant for deer overall (F= 2.716; df= 3, 52; P= 0.054) with dawn having a higher mean number of detections per hour ($\bar{x}$= 0.117 ± 0.022 SE) than day ($\bar{x}$ = 0.045 ± 0.012 SE) (Fig. 1). The treatment*time period interaction was not significant for deer detections overall (F= 0.850; df= 3,52; P=0.473) (Fig. 1A). Counter to our predictions, males were detected more frequently in hunted areas ($\bar{x}$= 0.036 ± 0.007 SE detections/hour) than in refuges ($\bar{x}$= 0.016 ± 0.005 SE detections/hour) (F= 4.616; df= 1, 52; P= 0.036) (Fig. 1B). For male deer, time period (F= 1.032; df= 3,52; P= 0.386) and the treatment*time period interaction was not significant (F= 1.337; df= 3,52; P= 0.273) (Fig. 1B). Female deer were detected more frequently during dawn ($\bar{x}$= 0.0690 ± 0.016 SE detections/hour) than day ($\bar{x}$= 0.0249 ± 0.008 SE detections/hour) (Fig. 1C), though the statistical significance was marginal (F= 3.080; df= 3, 52; P= 0.050). Treatment (F= 0.090; df= 1,52; P= 0.766) and the treatment*time period interaction was not statistically significant for females (F= 0.973; df= 3,52; P= 0.413) (Fig. 1C).

Discussion

Detection rates of deer were not higher in refuge areas than in hunted areas, regardless of sex, which does not support the hypothesis that deer avoid areas with hunting pressure. Contrary to our prediction, males were more active in hunted areas than in refuges. We hypothesized that deer would become more nocturnal in hunted areas in order to avoid hunting pressure; however, deer did not show increased levels of nocturnal activity in hunted areas. Deer were more active at dawn than any other time period, apparently driven by

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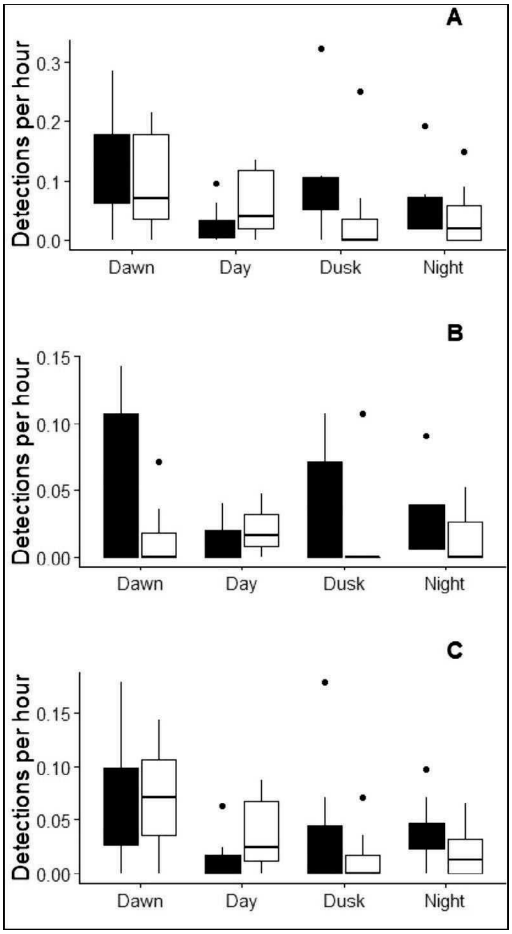


Figure 1. Mean detections per hour of white-tailed deer across time (periods; see text for time spans) between hunted (solid boxes) and refuge areas (open boxes) for (A) sexes pooled, (B) males only, and (C) females only. Boxplots display the median, upper and lower hinges which correspond to the first and third quartiles, as well as the upper whisker which extends from the upper hinge to the largest value no further than 1.5 times the inter-quartile range, and the lower whisker which extends from the lower hinge to the smallest value no further than 1.5 times the inter-quartile range. Points beyond the whiskers are considered outlying points.

the activity of females. Deer are known to be crepuscular, with higher levels of activity during dawn and dusk (Holzenbein and Schwede 1989; Schwartz and Schwartz 2016) than day.

Why did we not find patterns consistent with deer avoidance of hunted areas and peak times of hunting activity? It is possible that male deer were more active in hunted areas if there were more females to pursue there for mating; however, we found no significant pattern in female activity in vs. outside refuges to suggest this. Perhaps changes in behavior due to reproductive activities of white-tailed deer lessened the impacts of human hunting activity? In white-tailed deer, peak reproductive activity occurs during the second half of October and the first half of November (Smith 1991). During this time, both male and female deer become more active and cover larger distances daily (Holzenbein and Schwede 1989). Given that our study occurred during the rutting season it is possible that the lack of differences in activity between hunted and refuge areas was due to the increased amount of activity in both sexes during this time of the year.

Perhaps refuges were insufficiently sized to buffer against signals of human hunting activity. White-tailed deer are dispersive, especially during the rutting season, and occupy winter home ranges of around 600 ha (Wood, Mackie, and Hamlin 1989; Hygnstrom et al. 2008). These home ranges are large relative to the 520 ha to 2500 ha refuges included in our study. If refuges are not large enough to completely contain the home range of deer, it is possible that the deer in our study regularly encountered signals of hunting activity via human scent, gunfire, fleeing conspecifics, or movement of hunters (de Boer et al. 2004), irrespective of time spent within the refuges or their proximity to refuge borders when inside refuges. If deer are influenced by hunting activity both within refuges and hunted areas, then it is unlikely that deer would show a preference for refuges.

Another explanation as to why deer did not show the predicted differences in locations and periods of activity is that hunting pressure within hunted areas was insufficient to alter the

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Table 1. Percent coverage of landcover types on hunted areas and refuges within the Flint Hills National Wildlife Refuge (FHNWR) and Melvern Wildlife Area (MWA). Data were obtained from the 2015 Level I Kansas Land Cover Patterns layer (https://kansasgis.org/) and calculated using ArcGIS v 10.6 (ESRI, Redlands, CA).

	MWA Hunted	MWA Refuge	FHNWR Hunted	FHNWR Refuge
% Cropland	23	4	17	13
% Grassland	42	31	37	32
% Woodland	26	21	15	29
% Water	9	45	30	25

distribution of deer on the areas we studied. Root, Fritzell and Giessman (1988) determined that deer behaviors were impacted by human hunting pressure, but human hunting pressure was at the rate of 1.31 h of activity per ha per day. Autry (1967) concluded that in order for human activity to be "harassing" to deer populations, then the rate of human activity must be above 0.389 h of activity per ha per day. While the FHNWR does not collect data pertaining to hunter effort, the MWA uses an online "check-in" system (https://kdwpt.isportsman.net/) to monitor hunter activity and effort on the area (B. Niemann, Kansas Department of Wildlife, Parks and Tourism, pers. comm.). During our study, on the MWA, human hunting pressure occurred at a rate of 0.065 h of activity per ha, per day, which is less than the influential levels of hunting pressure suggested by Root, Fritzell and Giessman (1988) and Autry (1967).

Despite standardization of habitat in the immediate vicinity of camera traps, differences in landscape configuration between refuge and hunted areas might also have affected our results. Hunted areas on the MWA and FHNWR contain a greater proportion of croplands and grasslands than the refuge areas (Table 1). Also, the refuge on the MWA is covered primarily by water (Table 1). Deer have been found to frequent crop fields for foraging (Lesage et al. 2000) and areas of unharvested crops remain as wildlife food plots at FHNWR and MWA. If residing within hunted areas allow deer to access a greater quantity of food than in refuges, then predation risk from hunting may be negated by an increased abundance of foraging opportunities (Brown, Laundre and Gurung 1999).

In conclusion, white-tailed deer showed no changes in overall activity, or temporal periods of activity, between refuge and hunted areas. However, various factors might have masked the effects of human hunting activity on deer behavior. Additionally, improvements to our study could include a larger sample size of cameras and wildlife areas with refuges, which would improve statistical power and expand the geographic scope of inference. It is still possible that, according to previous research, that human hunting is altering white-tailed deer behavior and activity in Kansas. If land managers want to create refuge areas that deer can use to avoid hunting pressure, changes might need to be made in order to accomplish this goal. Refuge size might need to be increased in order to effectively encompass the home ranges of many white-tailed deer. Also, habitat and food resources within refuge areas might need to be managed so that they are similar to hunted areas.

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Acknowledgments

We would like to thank the Emporia State University Department of Biological Sciences for access to equipment used in the field. Our study would not have been possible without permission to access natural areas managed by the Kansas Department of Wildlife, Parks, and Tourism as well as the United States Fish and Wildlife Service and specifically Brad Niemann at the Melvern Wildlife Area and Jack Bohannan at the Flint Hills National Wildlife Refuge. We would also like to thank Erika Martin for her help in creating figures.

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